

Extra exercises of Lecture 3

EX 1

Consider an annual coupon bond with a duration of 2 years. If the interest rate, which currently is 15%, raises to 16%, what will be the approximate percentage change in the bond's price?

- a) The price will go down by 0.91%
- b) The price will go down by 1.74%
- c) The price will go up by 0.91%
- d) The price will go down by 1.65%

EX 2

The price of a 5-year zero coupon bond with face value of \$1,000 is \$903. The current interest rate is 1.5%. What is the bond's duration?

- a) 1.82
- b) 2.21
- c) 5
- d) Not enough information

EX 3

If you invest in a government bond whose time to maturity exceeds your investment horizon, which primary risk do you expose yourself to?

- a) interest rate risk
- b) inflation risk
- c) reinvestment risk
- d) default risk

EX 4

Compare two bonds: Bond A is an annual coupon bond of \$1,200 face value and of four years remaining until maturity, with 6% YTM and a market price of \$1,500. Bond B is a zero coupon bond and has two years remaining to maturity. Then:

- I. Bond B has a shorter duration than bond A
- II. Bond A has a shorter duration than bond B
- III. Bond B has shorter duration than bond A if and only if the YTM on bond B is below 6%
- IV. It cannot be determined